DTC P1794:00 [i-ELOOP]

id130201001700

Note
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".
- If the capacitor (i-ELOOP) voltage is low after replacing the DC-DC converter (i-ELOOP), the i-ELOOP indicator light (green) flashes until the capacitor (i-ELOOP) voltage is **approx. 9.5 V**.

Details On DTCs

DESCRIPTION	Capacitor (i-ELOOP) system: Capacitor (i-ELOOP) malfunction	
DETECTION CONDITION	Determination conditions	Capacitor (i-ELOOP) malfunction determined • Any one of the following conditions is met: — Capacitor (i-ELOOP) interrupt relay has a malfunction. — Capacitor (i-ELOOP) internal resistance exceeds the specification. — Capacitor (i-ELOOP) capacitance exceeds the specification. — Capacitor (i-ELOOP) voltage exceeds the specification. — Service plug is pulled out during vehicle travel. Capacitor (i-ELOOP) deterioration determined • Any one of the following conditions is met: — Capacitor (i-ELOOP) internal resistance exceeds the specification. — Capacitor (i-ELOOP) capacitance is less than the specification. Over-charge determined • Capacitor (i-ELOOP) voltage exceeds the specification.
	Preconditions	• Ignition switched ON
	Drive cycle	• 1
	Self test type	• CMDTC self test, ODDTC self test
	Sensor used	• DC-DC converter (i-ELOOP)
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function. (With i-stop) • Inhibits the i-ELOOP control. • Switches to bypass mode which shuts the capacitor (i-ELOOP) off.	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). (With i-stop) • Illuminates i-ELOOP warning light (amber). • The following vehicle conditions differ depending on the type of malfunction — Charging system warning light turns on (DC-DC converter (i-ELOOP) turns on during over-charge determination) — Master warning light turns on (illumination during bypass mode by i-ELOOP)	
POSSIBLE CAUSE	• Engagement malfunction in capacitor (i-ELOOP), DC-DC converter (i-ELOOP) • Capacitor (i-ELOOP) malfunction — Capacitor (i-ELOOP) deterioration — Capacitor (i-ELOOP) interrupt relay malfunction • Generator overcharge (in this case, PCM detects DTC P2504:00) • DC-DC converter (i-ELOOP) malfunction	

System Wiring Diagram
- Not applicable

Function Explanation (DTC Detection Outline)
- The capacitor (i-ELOOP) can store generated electrical energy from the generator efficiently.
- The electrical energy stored during deceleration fuel-cut is used during driving and while the engine is stopped by i-stop, which reduces the power generating load on the generator and the amount of fuel consumption.
- If a malfunction occurs with the capacitor (i-ELOOP), constant charging takes place by operation of the bypass mode which could cause the fuel economy to worsen.
- In this diagnostic, the internal resistance/capacitance is calculated from the capacitor (i-ELOOP) voltage value and charge/discharge current value, and the DC-DC converter (i-ELOOP) determines that there is a malfunction in the battery-related parts by the internal resistance/capacitance/capacitor (i-ELOOP) voltage exceeding the specified value, and it stores a DTC.

Repeatability Verification Procedure
1. If the negative battery cable is disconnected and re-connected, perform the battery condition initial setting (i-stop setting) to complete the i-stop initial setting. (See BATTERY CONDITION INITIALIZATION SETTING (i-stop SETTING).)
2. Start the engine.(no electrical load)
3. Depress the accelerator pedal to increase the engine speed to **3,000 rpm**.

4. Release the accelerator pedal to idle the engine.
5. Stop the engine.
6. Switch the ignition on and wait for **1 min**.
7. Switch the ignition off and wait for **20 min**.
8. Start the engine.(no electrical load)
9. Verify that Power Output (Normal) is displayed for the DC-DC converter (i-ELOOP) voltage reduction circuit operation status (PID: CONV_STAT) using the M-MDS data monitor function. (See PID/DATA MONITOR INSPECTION [i-ELOOP].)
   — If Power Output (Normal) cannot be verified, verify that DTCs related to the capacitor are detected. (See DTC INSPECTION [i-ELOOP].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
   - If DTCs are detected, repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].) (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
   - If DTCs are not detected, wait for 30 min or more and then implement Step 9 in the repeatability verification procedure again.

PID Item/Simulation Item Used In Diagnosis

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
CAP_SYS_10	Open circuit condition of the capacitor circuit	Off/On	• Capacitor (i-ELOOP) open circuit: On • Capacitor circuit is normal: Off
CAP_SYS_14	Capacitor (i-ELOOP) capacitance deterioration	Off/On	• Capacitor (i-ELOOP) capacitance status deteriorates: On • Capacitor (i-ELOOP) capacitance status is normal: Off
CAP_SYS_16	Capacitor (i-ELOOP) internal resistance deterioration	Off/On	• Internal resistance status in capacitor (i-ELOOP) deteriorates: On • Internal resistance status in capacitor (i-ELOOP) is normal: Off
CAP_SYS_17	Short circuit to ground malfunction in the capacitor (i-ELOOP)	Off/On	• Short circuit to ground in capacitor (i-ELOOP): On • No short circuit to ground in capacitor (i-ELOOP): Off

Function Inspection Using M-MDS

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: VERIFY IF DIAGNOSTIC RESULT IS AFFECTED BY DTC RELATED TO GENERATOR OVERCHARGE** • Switch the ignition off, then ON (engine off). • Perform the PCM DTC inspection using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the DTC P2504:00 also present?	Yes	Go to the applicable DTC inspection. (See DTC P2504:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC P2504:00 [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**PURPOSE: VERIFY IF POWER SUPPLY IS AFFECTED BY DTC RELATED TO DC-DC CONVERTER (i-ELOOP)** • Perform the DC-DC converter (i-ELOOP) DTC inspection using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs present?	Yes	DTC U3000:49 is displayed: • Go to the next step. Any other DTCs are displayed: • Go to the applicable DTC inspection. (See DTC TABLE [i-ELOOP].)
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.
4	**VERIFY THE OPEN CIRCUIT CONDITION OF THE CAPACITOR CUIRCUIT** • Access the CAP_SYS10 PID using the M-MDS. (See PID/DATA MONITOR INSPECTION [i-ELOOP].) • Is the CAP_SYS10 PID value off?	Yes	Reconnect the service plug, then go to the next Step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
5	**PURPOSE: DETERMINE IF THE MALFUNCTION IS CAUSED BY CAPACITOR (i-ELOOP) BY VERIFYING THE PID RELATED TO THE CAPACITOR (i-ELOOP)** • Access the following DC-DC converter (i-ELOOP) PIDs using the M-MDS (See PID/DATA MONITOR INSPECTION [i-ELOOP].) — CAP_SYS_10 — CAP_SYS_14 — CAP_SYS_16 — CAP_SYS_17 • Are all of the PID item “CAP_SYS10, CAP_SYS_14, CAP_SYS_16, CAP_SYS_17” values off?	Yes	Replace the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) Go to the troubleshooting procedure to perform the procedure from Step 2.
		No	Replace the capacitor (i-ELOOP). (See CAPACITOR (i-ELOOP) REMOVAL/ INSTALLATION [i-ELOOP].) Go to the troubleshooting procedure to perform the procedure from Step 2.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1
  - — Verify the engagement condition of each part.
- Step 2—3
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY IF MALFUNCTION CAUSED BY ENGAGEMENT MALFUNCTION OF EACH PART** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the service plug. (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Verify that the following parts are engaged correctly. — DC-DC converter (i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) — Capacitor (i-ELOOP) (See CAPACITOR (i-ELOOP) REMOVAL/ INSTALLATION [i-ELOOP].) • Is the engagement condition of each part normal?	Yes	Go to the next step.
		No	Tighten the DC-DC converter (i-ELOOP) or capacitor (i-ELOOP) properly, then go to the next step.

STEP	INSPECTION	RESULTS	ACTION
2	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Start the engine and leave for **100 s or more**. • Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Is the same DTC present?	Yes	Replace the capacitor (i-ELOOP) and verify that the repairs have been completed in Step 2. (See CAPACITOR (i-ELOOP) REMOVAL/ INSTALLATION [i-ELOOP].) • If the malfunction recurs, replace the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) — Start the engine and leave for **100 s or more**. — Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) Go to the next step.
		No	Go to the next step.
3	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [i-ELOOP].)
		No	DTC troubleshooting completed.